PRACTICAL JOURNAL

Commands only • Compiled from the uploaded 53-page journal

Note: Practical 2 and Practical 8 contain command screenshots in the source PDF that could not be read as text. Common command formats are included for Practical 2 and are marked as a note where exact commands were unavailable.

PRACTICAL 1 — USERS, GROUPS & PERMISSIONS

```
useradd username
passwd username
usermod -aG groupname username
whoami
last
userdel username
passwd -l username
passwd -u username
mkdir -p /data/girish
ls -ld directory
groupadd groupname
ls -ltr
usermod -G groupname username
ls -l
ls
chmod permissions filename
chgrp groupname filename
```

PRACTICAL 2 — RPM & YUM

```
# The source journal shows these commands only as screenshots, so they could not be extracted reliably.
rpm -ivh package.rpm
rpm -qa
rpm -qi package_name
rpm -Uvh package.rpm
rpm -e package_name
yum clean all
yum history
yum update
yum upgrade
yum remove package_name
```

PRACTICAL 3 — LVM LOGICAL VOLUME

```
fdisk -l
fdisk /dev/sda
pvcreate /dev/sda1
pvdisplay
vgcreate vgdata /dev/sda1
vgdisplay
lvcreate -L 500M -n lvdata vgdata
lvdisplay
mkfs.ext4 /dev/vgdata/lvdata
mkdir /mnt/lvdata
mount /dev/vgdata/lvdata /mnt/lvdata
df -h
vi /etc/fstab
```

```
/dev/vgdata/lvdata /mnt/lvdata ext4 defaults 0 0
```

PRACTICAL 4 — LVM SNAPSHOT

```
lvdisplay
lvcreate --size 2M --snapshot --name vol1_snapshot /dev/vgapps/app-lv
mkdir /mnt/vol1
mount /dev/vgapps/vol1_snapshot /mnt/vol1
ls /mnt/vol1
umount /mnt/vol1
lvremove /dev/vgapps/vol1_snapshot
lvdisplay
```

PRACTICAL 5 — SWAP FILE

```
fallocate -l 1G /swapfile
chmod 600 /swapfile
ls -lh /swapfile
mkswap /swapfile
swapon /swapfile
swapon --show
swapon -s
swapoff /swapfile
sysctl vm.swappiness=15
```

PRACTICAL 6 — SSH

```
service ssh start
service ssh status
ifconfig
ssh root@192.168.0.113 -o HostKeyAlgorithms=+ssh-rsa
ls
whoami
hostname
ssh root@192.168.0.113 -o HostKeyAlgorithms=+ssh-r
```

PRACTICAL 7 — POSTFIX MAIL SERVER

```
su -
cd /media/RHEL\ 6.0/i386/Disc\ 1/Packages
ls | grep samba
ls | grep postfix
rpm -ivh postfix-2.6.6-2.el6_i686.rpm
rpm -qa | grep postfix
rpm -qa | grep mailx
cd /etc/postfix/
cp main.cf main.bkp
hostname -f
vi main.cf
myhostname = localhost.localdomain
relayhost = [smtp.gmail.com]:587
mkdir /etc/postfix/sas1
cd /etc/postfix/sas1
touch sas1_passwd
vi sas1_passwd
postmap sas1_passwd
chmod 600 *
/etc/init.d/postfix start
```

```
/etc/init.d/postfix restart
service postfix start
service postfix status
echo "Test the mail practical by TYIT A student" | mail -s "tabish ansari" fmomin494@gmail.com
```

PRACTICAL 8 — DNS SERVER (BIND)

```
# The journal contains screenshots for this practical; no DNS command text was extractable.
:wq
```

PRACTICAL 9 — SAMBA FILE SHARING

```
df -Th
cd
cd Packages
ls | grep httpd
rpm -ivh samba-client
rpm -ivh samba-common
rpm -ivh samba-winbind
rpm -ivh samba-winbind-clients
rpm -qa
ifconfig
mkdir -p /home/TYIT
vi /etc/samba/smb.conf
chmod -R 777 /home/TYIT
getsebool -a
setsebool -P samba_enable_home_dirs=on
useradd student
passwd student
smbpasswd -a student
smbclient //10.0.2.15/home/TYIT -U student
smbclient //192.168.1.7/home/TYIT -U student
sudo vi /etc/samba/smb.conf
ls -ld /home/TYIT
sudo service smb restart
sudo service nmb restart
smbclient -L //192.168.191.130 -U student
```

PRACTICAL 10 — CRON JOB

```
rpm -qa | grep cron
service crond start
chkconfig crond on
service crond status
mkdir -p /root/cron_test
cd /root/cron_test
vi /root/cron_test/testscript.sh
#!/bin/bash
date >> /root/cron_test/testscript.txt
chmod +x /root/cron_test/testscript.sh
crontab -e
* * * * * /root/cron_test/testscript.sh
crontab -l
cat /root/cron_test/testscript.txt
```

PRACTICAL 11 — ONE-TIME TASK USING at

```
service atd status
```

```
service atd start
vi /tmp/at_test.sh
echo "At Job Executed on: $(date)" >> /tmp/at_log.txt
chmod +x /tmp/at_test.sh
at now + 5 minutes
/tmp/at_test.sh
# Press Ctrl+D to save the job
atq
at -l
cat /tmp/at_log.txt
```

PRACTICAL 12 — ANACRON

```
rpm -qa | grep cron
vi /etc/anacrontab
15 5 backup_trigger /tmp/backup.sh
anacron -f -d
vi /tmp/backup.sh
cat /tmp/backup.sh
ls -ltr
cd /var/log
ls cron
```